MSc in Computer Science and Engineering

Planning, Learning and Decision Making

Lab 2: Optimal Control in a Fragile Smart Home
Markov Decision Processes and Policy Analysis

1 Introduction

In Lab 1, you modeled the energy dynamics of a home as a passive Markov Chain. In this lab, you will take control. The goal is to minimize costs while preventing catastrophic system failure. You will study solution for Markov Decision Process, i.e. how to compute values and actions.

2 The MDP Environment

The state space remains the same as Lab 1: (B, W, C) plus a failure state (Index 75). However, transitions now depend on the chosen action $a \in \{AC, AS, AE\}$. However, transitions now depend on the chosen action $a \in \{AC, AS, AE\}$.

2.1 Transition Probability $P(s'|s, a)$

See the end of the document for details on the transitions (only relevant for question 4.)

Weather (W) and consumption (C) transitions are independent of the chosen action but are dependent on the starting weather conditions, meaning that at each step the distributions of W_{t+1} and C_{t+1} depend only on W_t (e.g. consumption is lower when the weather is sunnier).

Battery transitions are dependent on the selected action a. For $a = AS$, the distribution of the next battery level B_{t+1} also depends on the weather W_t (e.g. with sunnier weather the probability of successfully charging the battery increases). For $a = AC$ the charge happens deterministically and for $a = AE$ the charge reduces, with a small probability of transitioning to state 75.

The failure state always transitions to itself with probability 1, for all actions.

2.2 Cost Function $C(s, a)$

The cost is defined by the energy balance:

- **Grid Usage:** Each unit of energy bought from the grid costs 10.
- **Failure Penalty:** State 75 grants a cost of $R_{\text{fail}} = 500$.

- **Solar Revenue:** Selling excess solar grants -5.

You can access the file here

3 Questions (answers to be submitted)

1. How do you decide what is the best action to make at each instant? Moreover, if you have two policies, how do you select which one is better? The solution to each of these problems can be different or not.

2. Implement the value iteration (VI) and policy iteration (PI) algorithms. Show that the policies obtained by both methods display similar cost-to-go values. Which one converges faster? Why? What are the advantages and disadvantages of each one?
   Suggestion: Start by modeling the cost function $C(s, a)$. Additionally, track the learning. This can be the number of iterations, the Bellman Error: $||V_{k+1} - V_k||_\infty$, or the number of states where the optimal policy changed.

3. What is the impact of the cost?
   Suggestion: Run the value iteration with different cost values and see if the convergence rate, policy, mean-time between error changes. Additionally, give a high-level description of all cost functions implemented and add to the answer the following entries from every cost function: $C(0, :)$, $C(5, :)$, $C(32, :)$, $C(49, :)$, $C(75, :)$.

4. What is the importance of having the weather information in the state? Will it change the results? Why?
   Suggestion: Create a new transition matrix without the weather information, i.e. marginalized over those factors.

A Transition Probability $P(s'|s, a)$

Battery transitions are dependent on the selected action a. After any action a, probabilistic transitions for the environment (W, C) change following always the same distribution.

A.0.1 Battery B transitions

- **Action 0 (Charge - AC):**
  - 1.0 probability of normal charge: $B_{t+1} = \min(\max(B_t - C_t, 0) + 1, 4)$.
- **Action 1 (Solar charge - AS)**:

  This action charges from solar. The probability of charging p_{charge} depends on the current weather (W_t):
  - If $W_t = 0$: $p_{\text{charge}} = 0.05$, with $B_{t+1} = \min(\max(B_t - C_t, 0) + 1, 4)$;
  - If $W_t = 1$: $p_{\text{charge}} = 0.2$, with $B_{t+1} = \min(\max(B_t - C_t, 0) + 1, 4)$;
  - If $W_t = 2$: $p_{\text{charge}} = 0.5$, with $B_{t+1} = \min(\max(B_t - C_t, 0) + 1, 4)$;
  - If $W_t = 3$: $p_{\text{charge}} = 0.8$, with $B_{t+1} = \min(\max(B_t - C_t, 0) + 1, 4)$;
  - If $W_t = 4$: $p_{\text{charge}} = 0.95$, with $B_{t+1} = \min(\max(B_t - C_t, 0) + 1, 4)$.

  The remaining of the probability $1 - p_{\text{charge}}$ will be divided into a probability of transitioning to the failure state p_{fail}, and another probability, $p_{\text{discharge}}$, that accounts the reduction of the battery level due to the current consumption level:
  - If $\max(B_t - C_t, 0) = 0$, then $p_{\text{fail}} = 0.02$, otherwise $p_{\text{fail}} = 0$. With p_{fail}, the current state transitions to state 75.
  - If charging and failure do not occur, with $p_{\text{discharge}} = 1 - p_{\text{charge}} - p_{\text{fail}}$, we have that $B_{t+1} = \max(B_t - C_t, 0)$.
- **Action 2 (Export - AE)**:

  Sell solar to grid (1 battery level), use battery for home:
  - If $\max(B_t - C_t - 1, 0) = 0$, then $p_{\text{fail}} = 0.05$, otherwise $p_{\text{fail}} = 0$. With p_{fail}, the current state transitions to state 75.
  - If failure does not occur, with $p_{\text{discharge}} = 1 - p_{\text{fail}}$, we have that $B_{t+1} = \max(B_t - C_t - 1, 0)$.

A.0.2 Weather (W) transitions

- From $W_t = 0$: $P(W_{t+1} = 1) = 0.3$, $P(W_{t+1} = 0) = 0.7$.
- From $W_t = 1$: $P(W_{t+1} = 2) = 0.4$, $P(W_{t+1} = 0) = 0.3$, $P(W_{t+1} = 1) = 0.3$.
- From $W_t = 2$: $P(W_{t+1} = 3) = 0.4$, $P(W_{t+1} = 2) = 0.3$, $P(W_{t+1} = 1) = 0.3$.
- From $W_t = 3$: $P(W_{t+1} = 4) = 0.4$, $P(W_{t+1} = 3) = 0.3$, $P(W_{t+1} = 2) = 0.3$.
- From $W_t = 4$: $P(W_{t+1} = 4) = 0.7$, $P(W_{t+1} = 3) = 0.3$.

A.0.3 Consumption (C) transitions

- If $W_{t+1} \in \{0, 1\}$: C_{t+1} is uniformly distributed between 1 and 2 (inclusive), i.e., $P(C_{t+1} = c) = \frac{1}{2}, \forall c \in \{1, 2\}$.
- If $W_{t+1} = 2$: C_{t+1} is uniformly distributed between 0 and 2 (inclusive), i.e., $P(C_{t+1} = c) = \frac{1}{3}, \forall c \in \{0, 1, 2\}$.
- If $W_{t+1} \in \{3, 4\}$: C_{t+1} is uniformly distributed between 0 and 1 (inclusive), i.e., $P(C_{t+1} = c) = \frac{1}{2}, \forall c \in \{0, 1\}$.

A.0.4 Failure state (Index 75)

The failure state always transitions to itself with probability 1, for all actions.